Fiscal Year:	FY 2014	**Task Last Updated:**	FY 12/12/2014
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Project Title:	Center of Acute Radiation Research		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Radiation Effects Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19104-6072	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 NSBRI-RFA-08-02 Center of Acute Radiation Research (CARR)
Start Date:	09/01/2008	**End Date:**	08/31/2014
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2014 per December 2014 NSBRI report submission (Ed., 12/15/14) NOTE: End date changed to 4/30/2014 per December 2013 NSBRI report submission; Acute03 gap deleted per IRP Rev E (Ed., 1/8/14) Note: title change per NSBRI (12/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Freund, Gregory (University of Illinois at Urbana-Champaign) Wan, X. (University of Pennsylvania) Weissman, Drew (University of Pennsylvania) Cengel, Keith M.D., Ph.D. (University of Pennsylvania) Ebert, Douglas (Wyle Laboratories)		
Grant/Contract No.:	NCC 9-58-RE01801		
Performance Goal No.:			
Performance Goal Text:			

In this research funded by the National Space Biomedical Research Institute (NSBRI) called the Center of Acute Radiation Research (CARR) grant, animal model systems were utilized to evaluate and determine mechanisms involved in the space radiation induced acute radiation syndrome (ARS) with vomiting (and/or retching) and white blood cell counts in ferrets; white blood cell counts, fatigue, and immune system parameters in mice; and skin injury, with accompanying immune system changes, and white blood cell counts in pigs as the biological endpoints. In addition, the effects of combined exposure to simulated hypogravity and space radiation on blood cell counts and immune functions with respect to both the innate immune system and the acquired immune system were evaluated in mice. The first few years of the project were focused on determinations of whether certain parts of the ARS occur in animals exposed to space radiation(s) simulating that occurring during a solar particle event (SPE). If adverse biological effects were observed in these studies, later years of the project were focused on the development of countermeasures for these effects. Several drugs were evaluated in the animal model systems used to determine their effectiveness in prevention and/or mitigation of the specific ARS symptoms and signs.
The key findings of the first 5 years of the NSBRI CARR grant were reported in the 2013 annual progress report via the NSBRI website and the NASA Taskbook. During this past year of "cost extension" funding, the overall aims of the CARR cost extension work were as follows: Aim 1: To perform a limited series of new experiments in mice and rats that were targeted at critical areas in which our current experimental data and models remain incompletely understood. Aim 2: To continue analysis of data and banked tissue/blood samples from previous CARR experiments and perform exploratory analysis of existing data from human exposures to improve our ability to make specific recommendations for the prediction and management of human SPE radiation exposure toxicity.

The results of new experiments in mice led to the following conclusions:

1. Male or female mice of ages 11-12 months or 6 weeks old were treated with SPE proton radiation (2 Gy) and hindlimb suspension (R + S) as well as a bacterial challenge (Pseudomonas aeruginosa). Similar increases in morbidity and granulocyte counts were observed in the older male or female mice for radiation treatment

	alone and R + S treatments when compared to the results from 6 week old mice following a bacterial challenge. It is concluded that the combination of R + S has the same effect (resulting in a reduced ability to control a bacterial infection) in older (middle aged) male and female mice as observed for young (6 week old) mice.
Task Description:	2. The effects of reduced radiation doses on the ability of R + S mice to control a challenge with P. aeruginosa were evaluated. In these studies, it was observed that there was a significant increase in morbidity for doses of 1.5 and 2.0 Gy, but not for a dose of 1 Gy. These results demonstrate that in a murine model of spaceflight and SPE radiation exposure, a critical level of radiation, greater than 1 Gy, is needed to impair immune function resulting in an inability to control an otherwise controllable infection. 3. Experiments were performed to determine the efficacy of an antibiotic or granulocyte colony stimulating factor (G-CSF) in R + S treated C3H HeN mice and challenged with P. aeruginosa. An antibiotic (10 mg/kg enrofloxacin, q12 hr.) or granulocyte colony stimulating factor (G-CSF; pegfilgrastim, 600 microgram/kg, q3 days) led to a dramatic reduction in morbidity and bacterial load. Experimental work in rats involved exposure to a dose of 2 Gy SPE-like proton radiation with or without hindlimb suspension, both with and without an additional stressor of spaceflight (elevated carbon dioxide levels like those found in a spacecraft - approximately 1%, or 10,000 ppm, compared to normal levels [~0.03%]). The results indicated that exposure to 2 Gy SPE-like proton radiation resulted in statistically significant reductions in the numbers of white blood cells and lymphocytes during the first week post-irradiation, while the effects with or without additional stressors of spaceflight (hindlimb suspension and/or elevated carbon dioxide levels) were much like those observed in rats exposed only to the radiation dose. The results of studies performed to determine the effects of solar particle event (SPE)-like proton radiation in mice, ferrets, and Yucatan minipigs, on the numbers of circulating hematopoietic cells have been compared, through the use of relative biological effectiveness (RBE) values, in an attempt to extrapolate the risks from these animal model systems to astronauts exposed to SPE radiation. In mice, the RBE values for SPE-like protons (compared to conventional [gamma] radiation), did not differ significantly from 1. The RBE values for white blood cells (WBC) and neutrophil counts varied greatly between mice, ferrets, and pigs, with the RBE values being higher in ferrets than in mice, and considerably higher in pigs than in ferrets and mice. Based on these results, a mathematical model for the effects of SPE-like protons on WBC and neutrophil counts was developed. Using this model, the extrapolated results for humans were close to the values observed in minipigs, which suggests that studies in minipigs on hematopoietic cell counts could be more applicable to the results expected from astronaut exposures to SPE-like proton radiation than results from small animal studies. The results of other methods to extrapolate the results of previous CARR studies to astronauts suggest that for both superficial and deep organ systems, three specific factors, namely, individual astronaut anatomy, SPE type (particle/fluence spectrum), and shielding type (extravehicular activity (EVA) vs spacecraft) need to be modeled accurately in order to provide timely and evidence guided medical interventions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	It is well-known that exposure to ionizing radiation at sufficiently high doses can result in various types of adverse biological effects. The countermeasures to be developed by this research program have significance for all individuals exposed to high doses of ionizing radiation on Earth as well as in space. The countermeasures shown to protect against radiation induced biological effects from space radiation will be equally useful for protection against radiation induced adverse biological effects on Earth.
Task Progress:	The major aims were to perform a small number of animal experiments, and develop methods to predict the biological effects that could occur in astronauts exposed to solar particle event (SPE) radiation. The results from experiments in mice exposed to SPE-like proton radiation combined with simulated microgravity (hindlimb suspension [HS]) and a bacterial challenge (with Pseudomonas aeruginosa) were: 1) There was a significant increase in morbidity for mice exposed to doses of 1.5 and 2.0 Gy (but not 1.0 Gy), 2) Age (11-12 months of age vs 6 weeks old) and sex (male vs. female) do not alter the ability to control a bacterial infection, 3) An antibiotic (10 mg/kg enrofloxacin, q12 hr.) or granulocyte colony stimulating factor (G-CSF; pegfilgrastim, 600 microgram/kg, q3 days) led to a dramatic reduction in morbidity and bacterial load. Experimental rat work involved exposure to a dose of 2 Gy SPE-like proton radiation +/- HS, both +/- an additional stressor of spaceflight (elevated carbon dioxide levels like those found in a spacecraft -approximately 1%, or 10,000 ppm, compared to normal levels (~0.03%). The results indicated that exposure to 2 Gy SPE-like proton radiation resulted in statistically significant reductions in the numbers of white blood cells and lymphocytes during the first week post-irradiation, while the effects +/- the additional stressors of spaceflight were much like those observed in rats exposed only to the radiation dose. The results of previously performed studies to determine the SPE radiation effects in mice, ferrets, and minipigs on the numbers of circulating hematopoietic cells have been compared, through the use of relative biological effectiveness (RBE) values, in an attempt to extrapolate the risks from these animal model systems to astronauts exposed to SPE radiation. In mice, the RBE values for SPE-like protons did not differ significantly from 1. The RBE values for WBC and neutrophil counts varied greatly between mice, ferrets, and pigs, with the RBE values being higher in ferrets than in mice, and considerably higher in pigs than in ferrets and mice. A mathematical model for these effects was developed. Using this model, the extrapolated results for humans were close to the values observed in minipigs, suggesting that studies in minipigs on hematopoietic cell counts could be more applicable to the results expected from human exposures to SPE-like proton radiation than results from small animal studies. Other results related to the extrapolation of CARR animal studies to humans indicate that for both superficial and deep organ systems, three factors, namely, individual astronaut anatomy, SPE type (particle/fluence spectrum), and shielding type (EVA vs spacecraft) need to be modeled accurately in order to provide timely and evidence guided medical interventions.
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